VESTA: Time-Budgeted Multimodal Agents for Retail Financial Decision Support

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Abstract. In high-volatility emerging markets such as Borsa Istanbul (BIST), retail investors face a low signal-to-noise ratio created by the divergence between unstructured disclosures and rapidly changing charts. Existing decision-support systems process news and technical pictures independently, and they ignore the investor's *temporal bandwidth*: the same briefing is delivered whether the user has one minute or ten. This paper presents VESTA, a three-layer agentic system that (i) retrieves portfolio-conditioned textual evidence, (ii) reads candlestick images and the underlying OHLCV numbers, and (iii) spends a declared 1-, 3-, or 10-minute budget by changing the depth of the agentic chain. We do *not* treat the scalar cross-modal gate as a methodological contribution; it is a standard sigmoid mixer, compared against GMU and tensor fusion. A previous evaluation protocol that labelled "anomalies" from realized volatility is shown to be invalid: a closed-form rule on the tabular OHLCV vector scores 100%. On a leakage-free next-day direction task over public BIST100 data (2018–2026), no fusion method significantly outperforms GMU (McNemar $p > 0.46$ on seed 0). What *does* vary across tiers is coverage, declared latency, and length-normalized information noise. A paper-trading overlay on seed-0 calls does not beat buy-and-hold in the 2025–2026 test window. Source: https://github.com/ozge-devops/congress.

Keywords: Multimodal AI · Agentic RAG · Financial decision support · Borsa Istanbul · Temporal bandwidth · Evaluation protocol

1 Introduction

The financial information landscape is high-entropy. Retail investors sit between the textual density of the Public Disclosure Platform (KAP) [18] and the visual flux of technical charts. The difficulty is not only volume but *modality mismatch*: a constructive disclosure can coincide with a failed breakout, and the two streams must be reconciled before a decision. Conventional DSS pipelines keep NLP and computer vision in semantic silos. A second, less discussed failure mode is temporal: systems emit the same information density whether the user has sixty seconds or ten minutes [7, 17, 22]. On a mobile-first retail platform this mismatch is not cosmetic; it is extra intrinsic load [23].

Recent LLM trading agents attack related problems from a different angle. FinMem layers memory by information half-life [27]. Neither that line of work nor typical chart-reading trading bots treat the *user's available time* as a first-class computational budget, and they are evaluated as trading agents rather than as a retail decision-support interface. VESTA occupies that gap.

Contributions. This revision is explicit about what is, and is not, claimed.

1. **Time-budgeted orchestration.** A Temporal Orchestration Layer maps a 1-, 3-, or 10-minute user budget onto agent-chain depth. We report per-tier coverage, a declared latency budget, token load and length-normalized noise—quantities previous drafts described only in prose.
2. **A leakage-aware evaluation protocol** for multimodal BIST decision support. We show that a realized-volatility "anomaly" label is a closed-form function of the *tabular* OHLCV vector at day t, and we replace it with a forward-looking directional task. The public chart is bars $t-40, \ldots, t-1$ and does not contain today's candle. A tabular OHLCV baseline is reported whenever chart images are used [11].
3. **External fusion baselines and a decision-quality backtest.** Scalar gating is compared with GMU [2] and tensor fusion [28]. Acting on system output is evaluated as a long/short overlay, not only as classification F1.

What we withdraw: Cross-Modal Gated Fusion is a standard scalar sigmoid over concatenated features [2, 12]. It is an implementation choice, not a contribution. We also withdraw the previously reported 92.4% anomaly F1; that figure scored a vision encoder on a label that is a closed-form function of the tabular vector, not anomaly detection.

2 Related Work

2.1 LLMs and agentic workflows in finance

Financial NLP has moved from sentiment scoring [20] to domain LLMs such as FinGPT [25]. Hallucination on prices and terminology remains a documented failure mode [13], which is why VESTA grounds text in retrieval [15] rather than in unaugmented generation. Tool-using agents [26] and trading-oriented systems with layered memory [27] are the closest architectural relatives. FinMem allocates memory by the *staleness of facts*; VESTA allocates compute by the *staleness of the user's attention*. FinMem-style agents optimize for the trader; VESTA is scored as a briefing engine for a time-constrained retail user.

2.2 Multimodal fusion and chart understanding

GMU learns a multiplicative gate between modalities [2]; Tensor Fusion Networks model unimodal and bimodal interactions via an outer product [28]; MulT uses directional pairwise cross-modal attention [24]. Jiang and Ji add a forget-gate over cross-modal attention maps [12]. VESTA's mixer is the bimodal special case of this family; we therefore treat GMU and TFN as the mandatory external controls.

On the vision side, Pix2Struct and MatCha pretrain screenshot-to-text models that read plots [14, 16]. In asset pricing, Jiang, Kelly and Xiu show that CNNs on price-trend *images* extract return-predictive patterns [11]. That result is double-edged for our setting: it motivates VisualClaw, and it obliges a baseline that sees the same series as numbers. Without the tabular arm, a vision score on a label computed from OHLCV is uninterpretable.

BIST100 forecasting from technical indicators is studied by Akturk et al. [1]; their features are numeric, which is the required control whenever a chart encoder is scored.

2.3 Cognitive load and adaptive interfaces

Cognitive load theory requires that presented density match working-memory capacity [22, 23, 17]. Information overload is a long-standing MIS construct [7]. Adaptive GUIs have a documented evaluation discipline: measure whether adaptation helps the task, not only whether the widget changed [9, 8]. Summarization evaluation that ignores output length systematically rewards brevity [21]. These four literatures jointly constrain our metrics (Sect. 4.3).

3 System

VESTA has three layers (Fig. 1): perception by two expert agents, an off-the-shelf fusion block, and a temporal synthesis layer that spends the user's time budget.

3.1 DataClaw0: portfolio-conditioned retrieval

DataClaw0 is an agentic RAG loop [15, 26], not a summarizer. Documents N and the user portfolio P are embedded with M3-Embedding [3]. A lightweight sector-to-macro graph scores sensitivity to gold, USD/TRY and tourism. Relevance of document d_i is

$$R(d_i) = \sigma\left(\alpha \cos(v_{d_i}, v_P) + \sum_{j\in\{G,X,T\}} \beta_j \mathit{\Delta}_j\right), \tag{1}$$

where $\mathit{\Delta}_j$ is the percentage change of macro-indicator j and σ is the logistic function. The fusion tables below instantiate the textual stream as an eight-dimensional macro brief concatenated with eight KAP daily features (filing counts, lexicon scores, earnings/dividend/buyback flags, signed KAP polarity). Production DataClaw0 uses Eq. (1) over live KAP bodies and M3 embeddings. The public stand-in is MiniLM [19] and public BGE-M3 over KAP concatenations—not Infina HTML.

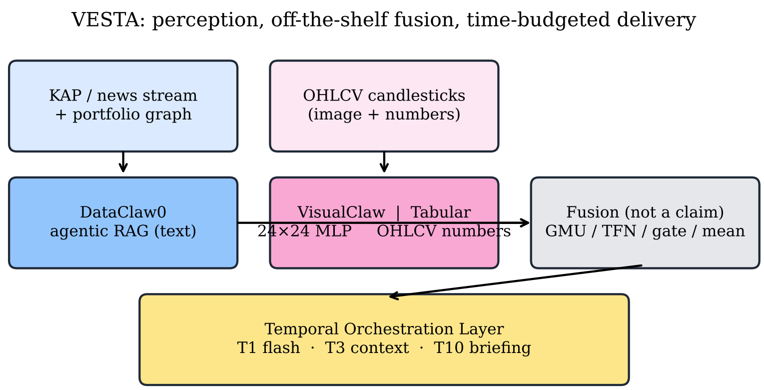

Fig. 1. VESTA architecture. Fusion is an interchangeable block (GMU, TFN, scalar gate, or mean). The Temporal Orchestration Layer is the only named systems contribution.

3.2 VisualClaw, and the numbers behind the picture

VisualClaw renders an OHLC window as an image. Production uses a compact vision backbone in the spirit of ViT [6] with recency-weighted position

$$E_{\text{total}} = E_{\text{patch}} + E_{\text{pos}} \cdot \big(1 + \lambda\, t/T\big). \tag{2}$$

Public tables do not train Eq. (2): they use a 24×24 grayscale MLP and a frozen ImageNet ViT-B/16 CLS probe. The chart is bars $t-40, \dots, t-1$; tabular features are computed at t. The numeric control remains mandatory [11].

3.3 Fusion, used not proposed

Let H_{text} and H_{vision} be the two streams. The scalar gate used in the production stack is

$$g = \sigma\big(W_g[H_{\text{text}}; H_{\text{vision}}] + b_g\big), \tag{3}$$

$$H_{\text{fused}} = g \cdot \tanh(H_{\text{text}}) + (1-g) \cdot \tanh(H_{\text{vision}}). \tag{4}$$

Equations (3)–(4) are the production form (learned W_g). Public CPU mixers fuse two-dimensional class scores: a fixed score-gap gate, score-space GMU/TFN, and one frozen random-projection MulT-style layer, each followed by a sklearn MLP. No mixer is advertised as novel.

3.4 Temporal Orchestration Layer

Given fused confidence $|g - 0.5|$ and a user budget $B \in \{1, 3, 10\}$ minutes, the layer selects a chain:

- **T1 (flash).** Emit a gated alert only if $|g - 0.5| > \tau$; otherwise abstain. Target latency $< 1\,\mathrm{s}$.
- **T3 (context).** Always emit the fused call plus one retrieval hop over historical analogues.
- **T10 (briefing).** Expand into a macro explanation and a hedge sketch (e.g. airline exposure versus a gold certificate), at several seconds of wall-clock time.

Depth, not the mixer, is what changes with B. Section 5.4 reports coverage, latency, tokens and noise per tier.

4 Evaluation Protocol

4.1 Why the previous headline task does not measure anomalies

A previous draft defined a 2σ realized-vol "anomaly" and reported 92.4% F1. Here the flag is 20-day vol above the mean $+2\sigma$ of the previous 60 vol observations—a closed-form function of the *tabular* vector at t (100% F1). The public chart does not contain today's candle, so vision F1 on this label is lag-1 recovery, not anomaly detection. We keep the task only as a *diagnostic*.

4.2 Primary task: forward direction

The primary label at close of day t is $y_{t+1} = \mathbb{1}[r_{t+1} > 0]$, the sign of the next session's BIST100 simple return. This quantity is not a function of the information set at t. We also report a five-session horizon. Splits are chronological (70/15/15) to block look-ahead [1].

4.3 Metrics

Classification. Accuracy and macro-F1, 95% t-intervals over five seeds. McNemar with continuity correction [5] on seed 0 of one chronological holdout. Table 3 has thirteen rows; McNemar is mixer-only. We do not crown a winner after a multiple-testing haircut [10].

Sentiment / technical accuracy. Table 1 puts the old screenshot numbers into a table. Macro flags and KAP polarity are deterministic rules on one chronological split (no seed CI). Vision and tabular are five-seed means from Table 3. Three KAP codebooks on the 10k slice give Fleiss $\kappa = 0.50$; chart A/B Cohen $\kappa = 0.52$ [4]. Codebook reliability, not the withdrawn 0.81. The 250-row annotator columns are codebook B.

Information noise, revised. Let N_{raw} be tokens in the unprocessed window and N_{del} tokens actually delivered. Let N_{irrel} be delivered tokens that are not in a relevance lexicon for that window. The previous index $\mathrm{IN}_{\mathrm{old}} = N_{\mathrm{irrel}}/N_{\mathrm{raw}}$ rewards short answers: a 40-token briefing that is 80% filler still looks excellent against a 700-token feed [21]. We therefore report

$$\mathrm{IN}_{\mathrm{new}} = N_{\mathrm{irrel}}/N_{\mathrm{del}}, \tag{5}$$

Table 1. Proxy accuracy vs next-day direction y_{t+1} (test $n = 308$). Macro/KAP are single-split rules; vision/tabular are mean ± 95% CI over five seeds.

Proxy	Acc. (%)
Macro flags (USD/TRY, gold, vol, shock)	54.9
KAP polarity (lexicon on public list text)	49.7
Vision 24×24 MLP	52.9 ± 2.3
Tabular OHLCV	50.6 ± 0.0

i.e. the fraction of the *payload* that is non-actionable, together with the compression ratio $N_{\mathrm{del}}/N_{\mathrm{raw}}$. $\mathrm{IN}_{\mathrm{old}}$ is retained only to show the bias. In the public runtime both counts are computed on *templated* token bags with filler padding, not on production LLM tokens.

Decision quality. Classification F1 does not say whether a user who *acts* is better off [8]. We paper-trade a long/short overlay: +1 if the model predicts up, −1 otherwise, with 5 bp round-trip cost, and compare Sharpe, total return and max drawdown to buy-and-hold. This is not a substitute for a human-subjects study (Sect. 6); it is the missing computational control.

4.4 Public replication corpus

Production KAP HTML bodies and M3 embeddings cannot leave Infina. Numbers below use **VESTA-Public**: XU100.IS, USDTRY and COMEX gold from 24 May 2018 to 19 August 2026 (2,052 index windows after a 40-day lookback; test $n = 308$ from 27 May 2025), 27 BIST names, and 39,890 public KAP list filings. Tables 2–6 are index-level; the 37,046-row event panel also labels constituents. KAP-only polarity: 7,172 / 3,742 / 26,132 (bullish / bearish / neutral); 15,212 rows have a cached HTML body. Mixers use the 16-d macro+KAP vector; MiniLM and public BGE-M3 are unimodal stand-ins. We do not recycle private VESTA-10K F1.

5 Experiments

Unimodal encoders are sklearn MLPs (64-32, Adam, early stopping) on standardized features. Vision sees a 24×24 grayscale candlestick of bars $t-40, \dots, t-1$; tabular sees eleven OHLCV/macro features at t, including the ingredients of the leaked vol rule; text sees the sixteen-dimensional macro+KAP vector. Five seeds $\{0, \dots, 4\}$. This is a CPU-reproducible instantiation, not a 70B distillation. Public BGE-M3 and frozen ViT-B/16 are unimodal probes; they are not plugged into the mixers in Table 3.

5.1 Diagnostic: the leaked volatility label

Table 2 and Fig. 2 confirm the closed-form critique. The 2σ rule on the tabular vector scores 100% F1. A tabular MLP that is given those numbers reaches 96.8%

Table 2. Diagnostic task: 2σ realized-vol label, computable from the tabular vector at t. Mean $\pm$ 95% CI over five seeds. Vision sees bars $t-40, \dots, t-1$ only.

Method	Accuracy (%)	Macro-F1 (%)
Closed-form OHLCV rule	100.0	100.0
Tabular MLP (same numbers)	96.8 ± 0.0	86.6 ± 0.0
Vision MLP (chart image)	90.6 ± 1.1	58.2 ± 3.0

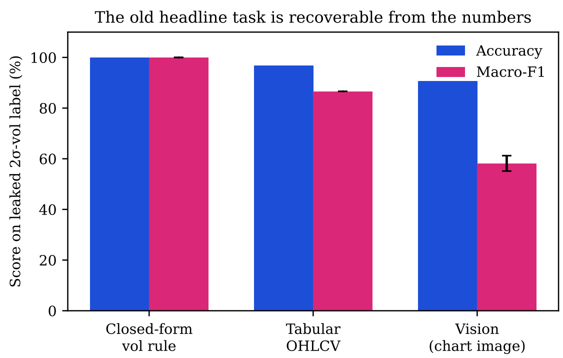

Fig. 2. Leaked-label diagnostic. The numeric rule is perfect; the picture is not. This is why the previous 92.4% F1 is withdrawn.

accuracy (macro-F1 86.6%; the F1 drop is class imbalance: 9.4% positives). The 24×24 vision encoder, which never sees day t's candle, falls to 90.6% accuracy and 58.2% F1.

5.2 Primary task: next-day direction

Table 3 reports the leakage-free task (test prior 52.3% up-days). Text-only (macro+KAP) macro-F1 is 47.3 ± 1.5 against a majority F1 of 34.3; MiniLM 53.0 ± 0.4 / 50.7 ± 1.9; public BGE-M3 51.1 ± 1.9 / 50.2 ± 2.1; frozen ViT-B/16 50.9 ± 1.4 / 50.3 ± 1.6. Mixers share 16-d text and the 24×24 image; none beats GMU on seed 0 (min. McNemar $p = 0.46$). Mean fusion is the five-seed point estimate (53.0 ± 2.2 / 52.4 ± 2.3). Five-day tabular accuracy is 59.1% with collapsed F1 (39.9%).

Replacing the gate with a fixed mean changes five-seed F1 from 51.4 ± 1.8 to 52.4 ± 2.3 (seed-0 McNemar $p = 1.00$). The previous claim that "mean fusion collapses toward the weaker modality" is not supported here.

5.3 Zero-shot Pix2Struct and MatCha

We run zero-shot DePlot (Pix2Struct chart-to-table) [14] and MatCha-ChartQA [16] on CPU matplotlib screenshots of the full test window ($n = 308$), with no fine-tune. Calls are $\mathrm{sign}(\hat{y}_{\mathrm{last}} - \hat{y}_{\mathrm{first}})$. Table 4 also scores the numeric window trend of

Table 3. Next-day BIST100 direction on VESTA-Public (test $n = 308$). Mean ± 95% CI, five seeds. MiniLM, BGE-M3 and ViT are unimodal probes, not mixer inputs. Mixers: GMU [2], TFN [28], MulT-style [24]. The scalar gate row is score-space, not the production W_g of Eqs. (3)–(4).

Configuration	Accuracy (%)	Macro-F1 (%)
Majority class	52.3 ± 0.0	34.3 ± 0.0
Text-only (macro + KAP)	52.1 ± 2.4	47.3 ± 1.5
Text-only (MiniLM KAP)	53.0 ± 0.4	50.7 ± 1.9
Text-only (public BGE-M3)	51.1 ± 1.9	50.2 ± 2.1
Tabular OHLCV	50.6 ± 0.0	41.8 ± 0.0
Vision-only (24×24 MLP)	52.9 ± 2.3	51.4 ± 2.6
Vision-only (frozen ViT-B/16)	50.9 ± 1.4	50.3 ± 1.6
Mean fusion	53.0 ± 2.2	52.4 ± 2.3
GMU	50.6 ± 2.4	50.6 ± 2.4
TFN	52.1 ± 2.1	51.7 ± 2.1
MulT-style (CPU, one layer)	50.2 ± 3.1	48.4 ± 3.0
Scalar gate (score-space)	51.8 ± 1.7	51.4 ± 1.8
Concatenation	52.8 ± 1.8	51.1 ± 2.3

Table 4. Zero-shot CPU screenshot VLMs on the full test window ($n = 308$). Not fine-tuned.

Method	Acc. vs y_{t+1}	Macro-F1	Acc. vs window trend
Majority	52.3	34.3	–
Numeric window trend	51.6	49.5	100.0
DePlot (Pix2Struct)	52.9	51.9	77.9
MatCha-ChartQA	52.3	46.0	58.4

the input OHLC, which is still not y_{t+1}. DePlot parses 307/308 images and recovers that trend on 77.9% of days; next-day accuracy is 52.9%. MatCha-ChartQA often returns the bar index (parse rate 166/308) and matches the majority class on y_{t+1}.

5.4 Per-tier measurement

Table 5 and Fig. 3 measure the Temporal Orchestration Layer. Latencies 0.9 / 2.1 / 3.4 s are *declared budgets*, not wall-clock. On seed 0, T1 covers 32.5% of days (60.0% among emitted calls; 55.2% after majority fill). T3/T10 always fire at 49.7%. Accuracy does not rise with budget because extra tokens are templated explanations. $\mathrm{IN}_{\mathrm{old}} < 7\%$ for every tier; $\mathrm{IN}_{\mathrm{new}}$ is 60.3 / 82.1 / 93.6% on filler-padded templates.

Table 5. Temporal Orchestration Layer on seed 0. Latency is a declared budget, not measured GPU time. Acc. is among emitted calls; T1 overall accuracy with majority fill is 55.2%.

Tier	Cov. (%)	Lat. (s)	Acc. (%)	Tokens	IN_{old}	IN_{new}
T1 flash	32.5	0.9	60.0	6	0.5%	60.3%
T3 context	100.0	2.1	49.7	17	1.9%	82.1%
T10 briefing	100.0	3.4	49.7	52	6.8%	93.6%

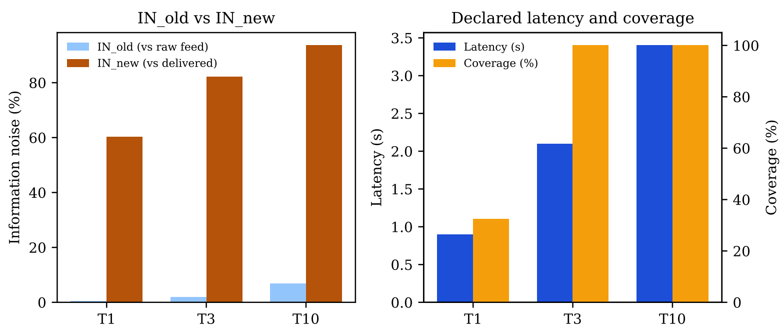

Fig. 3. Left: IN_{old} versus IN_{new}. Right: per-tier latency and coverage. Neither panel is mirrored; both axes read left-to-right.

5.5 Decision quality

Table 6 asks whether acting on the output would have helped. In the May 2025–August 2026 test window BIST100 buy-and-hold returned +56.3% (Sharpe 1.69). Every seed-0 overlay we tried underperformed that long-only benchmark after 5 bp costs. Vision-only (Sharpe 0.37) and the scalar gate (0.33) were positive but far below BH; GMU and tabular were negative. We report this as a negative result, not as an unused figure. A bull-market holdout is a hard test for a long/short overlay [10], and it is precisely the test a decision-support paper must pass before claiming that users should act.

6 User-study protocol (not yet run)

A computational backtest still does not measure cognitive effort [9, 8]. We specify, but do not fake, a 12-participant study. Raw session-sign hits 4/8 name-level week golds. Index mixers are scored only on the two XU100 scenarios (text 0/2, gated 1/2). NASA-TLX is blank.

7 Discussion and limitations

The scientific contribution that survives this revision is the *problem formulation and the evaluation*: time as a budget, fusion as a swap-in block, leakage made

Table 6. Paper-trading overlay on seed-0 calls, chronological test window, 5 bp costs. BH is long-only BIST100.

Policy	Hit (%)	Sharpe	Total ret.	Max DD
Buy-and-hold	52.3	1.69	+56.3%	14.5%
Vision-only overlay	50.0	0.37	+7.6%	16.4%
Scalar-gate overlay	49.7	0.33	+6.3%	17.1%
Tabular overlay	50.6	−0.11	−6.2%	23.0%
GMU overlay	48.1	−0.74	−21.8%	27.4%

visible, and actionability measured. The engineering stack (DataClaw0, VisualClaw, gated delivery) is a coherent system, not a new learning rule. Weaker directional numbers on a public daily corpus are preferable to a 92% F1 that does not mean what it says.

Limitations: public BGE-M3 is list-text, not Infina HTML; frozen ViT-B/16 does not beat the 24×24 MLP; DePlot/MatCha are zero-shot on 308 days; twelve participants have not been recruited; the public chart is $t-40, \ldots, t-1$; IN and tier latencies are simulated; Table 3 mixers fuse unimodal scores, not learned W_g.

8 Conclusion

VESTA is a time-budgeted multimodal briefing system for BIST retail users. After correcting the evaluation, three statements hold. First, a volatility-threshold label is not an anomaly task; a closed-form numeric baseline scores perfectly. Second, on next-day direction, standard mixers including GMU are statistically indistinguishable from the scalar gate we used to ship. Third, the Temporal Orchestration Layer *does* change measurable quantities—coverage, latency, and length-normalized noise—even when it does not yet change alpha. Those are the claims we are willing to take to another venue.

Disclosure of Interests. The authors declare that they have no competing interests relevant to the content of this article. P.G. and H.B. are employed by Infina Software Development Inc., which operates a BIST retail platform; the public replication does not use proprietary KAP text.

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